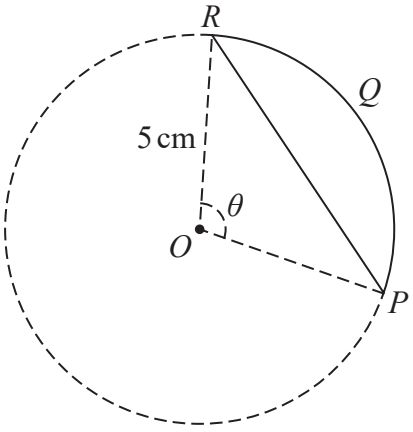



**Figure 1**

Figure 1 shows a sketch of a segment $PQRP$ of a circle with centre $O$ and radius 5 cm.

Given that

- angle $POR$ is $\theta$ radians
- $\theta$ is increasing, from 0 to $\pi$, at a constant rate of 0.1 radians per second
- the area of the segment $PQRP$ is $A$ cm$^2$

(a) show that

$$\frac{\mathrm{d}A}{\mathrm{d}\theta} = K\left(1 - \cos\theta\right)$$

where $K$ is a constant to be found.

**(2)**

(b) Find, in cm$^2$ s$^{-1}$, the rate of increase of the area of the segment when $\theta = \frac{\pi}{3}$

**(4)**